

1) Array in Javascript

Yes, in JavaScript, an array is a data structure used to store multiple values in a single variable. Arrays in JavaScript are ordered collections that allow you to store lists of items (like numbers, strings, or even other arrays) and access them by index. JavaScript arrays are flexible because they can store values of any type, and they can dynamically change in size.

Basic Array Syntax

You can create an array in JavaScript using either the array literal `[]` or the `Array` constructor:

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```
// Using array literal
let fruits = ["apple", "banana", "cherry"];

// Using Array constructor
let numbers = new Array(1, 2, 3, 4);
```

Accessing Elements

Elements in an array are accessed by their index, starting from `0`:

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```
console.log(fruits[0]); // Output: "apple"
console.log(numbers[2]); // Output: 3
```

Common Array Methods

JavaScript provides many methods for working with arrays, including:

- `.push()` – adds an element to the end of the array
- `.pop()` – removes the last element from the array
- `.shift()` – removes the first element from the array
- `.unshift()` – adds an element to the beginning of the array
- `.length` – returns the number of elements in the array

- `.map()` – applies a function to each element and returns a new array
- `.filter()` – returns a new array with elements that pass a test function
- `.forEach()` – runs a function on each element (without returning anything)

Example

Here's an example that shows how to work with an array in JavaScript:

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```
let numbers = [1, 2, 3, 4, 5];

// Add an element to the end
numbers.push(6); // [1, 2, 3, 4, 5, 6]

// Remove the first element
numbers.shift(); // [2, 3, 4, 5, 6]

// Double each number using map
let doubled = numbers.map(n => n * 2); // [4, 6, 8, 10, 12]

console.log(doubled);
```

JavaScript arrays are versatile, making them essential for handling collections of data. Let me know if you want more specific information about any array methods or operations!

2) Event Bubbling and Event Delegation

Event Bubbling and **Event Delegation** are core concepts in JavaScript, particularly when dealing with DOM events.

Event Bubbling:

- When an event occurs on an element, it first triggers the event handlers on that element.
- Then, the event "bubbles up" to its parent element, triggering any event handlers attached to the parent.
- This process continues up to the root of the DOM (document).

For example, if you click a button inside a `<div>`, the click event will first be handled by the button and then bubble up to the `<div>`.

Example:

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```
<div id="parent">
  <button id="child">Click Me</button>
</div>

<script>
  document.getElementById('child').addEventListener('click', () => {
    console.log('Child clicked');
  });

  document.getElementById('parent').addEventListener('click', () => {
    console.log('Parent clicked');
  });
</script>
```

If you click the button, the console will show:

 Copy code

```
Child clicked
Parent clicked
```

Event Delegation:

- Event delegation takes advantage of event bubbling by attaching a single event listener to a parent element.
- Instead of adding multiple event listeners to individual child elements, the event is handled by the parent.
- The event is captured during the bubbling phase, and based on the `event.target`, the specific child that triggered the event is identified.

This is useful for dynamically added elements since you don't need to bind an event listener to each new element individually.

Example:

 Copy code

```
<ul id="parent">
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
```

```
</ul>

<script>
  document.getElementById('parent').addEventListener('click', (event) =>
    if (event.target.tagName === 'LI') {
      console.log('Clicked on:', event.target.textContent);
    }
  });
</script>
```

If you click on an ``<li>``, it will log the clicked item's text. This way, even if you add new ``<li>`` items dynamically, the same event listener works for them without needing to add new listeners.

Benefits of Event Delegation:

- **Memory efficiency:** One event handler on a parent instead of multiple on individual child elements.
- **Dynamic elements:** It works seamlessly with dynamically added child elements.

Stopping Event Bubbling:

If you don't want an event to propagate (bubble) to parent elements, you can stop it using ``event.stopPropagation()``.

Example:

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```
document.getElementById('child').addEventListener('click', (event) => {
  event.stopPropagation(); // Stop bubbling
  console.log('Child clicked');
});
```

Now, only "Child clicked" will be logged, and the parent's event listener won't be triggered.

3) Event Capturing

Event Capturing (also known as **Event Tricking**) is another phase of how events are handled in the DOM. It occurs before **Event Bubbling** and follows the flow

from the root of the document down to the target element. Here's how it works compared to event bubbling:

Event Phases:

1. **Event Capturing (Trickling):** The event moves from the top of the DOM (document) down to the target element.
2. **Target:** The event reaches the target element where it was triggered.
3. **Event Bubbling:** After the event is handled at the target, it bubbles up through its parent elements back to the root of the DOM.

By default, most events do not use event capturing, but you can enable it by passing an additional argument to the `addEventListener` method.

Example of Event Capturing:

 Copy code

```
<div id="grandparent">
  <div id="parent">
    <button id="child">Click Me</button>
  </div>
</div>

<script>
  document.getElementById('grandparent').addEventListener('click', () => {
    console.log('Grandparent capturing');
  }, true); // Enable capturing phase

  document.getElementById('parent').addEventListener('click', () => {
    console.log('Parent capturing');
  }, true); // Enable capturing phase

  document.getElementById('child').addEventListener('click', () => {
    console.log('Child capturing');
  }, true); // Enable capturing phase
</script>
```

When the button is clicked, the event will flow downwards from the `grandparent` to `child` because capturing is enabled.

The console will log:

 Copy code

Grandparent capturing
Parent capturing
Child capturing

Capturing vs Bubbling:

- **Capturing:** The event moves from the outermost parent to the innermost target element.
- **Bubbling:** The event moves from the innermost target element back up to the outermost parent.

Using Event Capturing:

To use event capturing, you must pass `true` as the third argument of `addEventListener`:

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```
element.addEventListener('click', handler, true);
```

If you pass `false` (the default), the event listener will be executed during the bubbling phase instead:

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```
element.addEventListener('click', handler, false); // Default is false (
```

Practical Use Cases for Event Capturing:

Event capturing is less commonly used than event bubbling, but it can be useful in cases where you want to handle events at a higher level before they reach their target.

For example, if you have a form that is deeply nested and want to validate or prevent certain events before they hit specific form elements, event capturing can be useful.

4) First Order Function

A **first-order function** is a function that does not take other functions as arguments and does not return a function. It's a regular function that works with basic types such as numbers, strings, or objects.

Here's an example in JavaScript:

 Copy code

```
function add(a, b) {  
  return a + b;  
}  
  
const result = add(5, 3); // result is 8
```

In this case, `add` is a first-order function because it operates on the simple values `a` and `b` and does not take or return other functions.

In contrast, a **higher-order function** either takes another function as an argument, returns a function, or both.